

University of Luxembourg  
Master in Financial Mathematics

Optimization Techniques in Portfolio  
Management

Student Project 2

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## **Declaration of Authorship**

I declare that this report is my own work. All sources and materials used in the preparation of this report have been properly cited. I also confirm that I have not submitted this work, in whole or in part, for another assessment.

## **Use of Generative AI**

Generative AI tools were used as supporting tools during the preparation of this report. They were mainly used to check English grammar and wording, improve the clarity of some explanations, and help with technical issues related to LaTeX formatting and Python debugging. The mathematical content, choice of methods, examples, interpretation of the results, and final version of the report were reviewed and decided by the author.

# 1 Introduction

Portfolio optimization is an important part of portfolio management because it answers the question of how capital should be allocated across different assets. The main goal is to allocate capital in order to achieve certain performance objectives, depending on the investor's or portfolio manager's preferences.

In practice, financial data are complex and do not follow simple assumptions. Returns can change over time, show volatility clustering, and include extreme events. Because of this, classical models are not always sufficient. A useful portfolio optimization framework should be mathematically tractable and at the same time flexible enough to be applied to real financial data.

This report provides a survey of basic portfolio optimization techniques and compares their main ideas, strengths, and limitations in practical settings. It first introduces the basic notation and portfolio quantities. Then it presents the mean–variance framework as the starting point of modern portfolio theory. The report also discusses some limitations of the mean–variance model, including concentration risk, sensitivity to parameter estimation, and the need for practical constraints.

The next parts of the report consider alternative risk measures, practical constraints, and risk-based allocation ideas such as risk parity. Finally, an empirical out-of-sample backtest based on daily stock returns is used to compare different portfolios and to evaluate their practical behaviour.

This report is mainly based on selected chapters of Palomar's *Portfolio Optimization: Theory and Application*, especially Chapter 2 on financial data, Chapters 6 and 7 on portfolio basics and modern portfolio theory, Chapter 8 on portfolio backtesting, and Chapter 11 on risk parity portfolios [2].

# 2 Portfolio Basics

In this section, we introduce the main concepts and notation used in portfolio modelling.

Consider  $N \in \mathbb{N}$  financial assets and a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ . At each time  $t$ , let  $r_t : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}^N$  be the random vector of daily asset returns. Its component  $r_{i,t}$  denotes the daily return of asset  $i$  at time  $t$ .

The expected return vector and the covariance matrix of  $r_t$  are defined by

$$\mu := \mathbb{E}[r_t] \in \mathbb{R}^N, \quad \Sigma := \text{Cov}(r_t) \in \mathbb{R}^{N \times N}.$$

The matrix  $\Sigma$  represents the dependence of  $N$  random variables in random vector  $r_t$ . In traditional mean–variance models, it is also used as a proxy for the correlated risk of  $N$  interdependent assets.

A portfolio is represented by a weight vector  $w \in \mathbb{R}^N$ , where  $w_i$  denotes the proportion of total capital invested in asset  $i$ . The standard budget constraint is

$$\mathbf{1}^T w = 1,$$

where  $\mathbf{1}$  is the vector of ones.

The portfolio return at time  $t$  is the random variable

$$R_{p,t} := w^T r_t,$$

where the subscript  $p$  refers to the portfolio. Since the weights sum to one,  $R_{p,t}$  is a weighted average of the asset returns.

The expected return and variance of the portfolio are given by

$$\mathbb{E}[R_{p,t}] = w^T \mu, \quad \text{Var}(R_{p,t}) = w^T \Sigma w.$$

Here,  $w^T \Sigma w$  is the portfolio variance. In the mean–variance model, it is used as a proxy for portfolio risk, while volatility refers to the standard deviation of portfolio returns given by  $\sqrt{w^T \Sigma w}$ .

In practice, portfolio managers aim to achieve different objectives when constructing a portfolio. The typical goals are to maximize expected return, minimize risk, and build a well-diversified portfolio. To address these objectives, different types of allocation strategies are used.

Simple heuristic approaches are often applied because they are easy to implement. For example, the Global Maximum Return Portfolio (GMRP) focuses on maximizing expected return, but it may lead to a highly concentrated portfolio. In contrast, the equally weighted portfolio ( $1/N$ ) distributes capital equally across all assets and provides a simple diversified benchmark. The quintile portfolio selects assets based on ranking criteria and allocates weights to the top-performing group. Another group of methods focuses primarily on risk rather than expected return. These risk-based portfolios rely on the covariance matrix. The Global Minimum Variance Portfolio (GMVP) aims to minimize overall portfolio variance. Since the covariance matrix includes the dependence between asset returns, the GMVP can improve diversification when assets are not perfectly correlated. Similarly, the inverse volatility portfolio assigns higher weights to low-risk assets. More advanced approaches, such as risk parity and diversification-based portfolios, further improve allocation by explicitly controlling the contribution of each asset to total risk.

This section has introduced the basic notation and some portfolio construction ideas. These will be used in the next section, where we discuss the mean–variance portfolio model as the starting point of modern portfolio theory.

### 3 Modern Portfolio Theory

Modern Portfolio Theory (MPT) started with the mean–variance portfolio model introduced by Harry Markowitz in 1952 [1]. The Markowitz model is the first model in MPT that formulates portfolio optimization as a multi-objective problem. It balances two traditional objectives of portfolio managers: maximizing expected return and minimizing risk. This trade-off is the main idea behind the mean–variance portfolio model discussed in the following.

### 3.1 Mean–Variance Portfolio (Markowitz Model)

The mean–variance portfolio, also called the Markowitz model, is the basic model studied in this section.

Using the notation introduced in Section 2, let  $w \in \mathbb{R}^N$  denote the portfolio weight vector,  $\mu \in \mathbb{R}^N$  the vector of expected returns, and  $\Sigma \in \mathbb{R}^{N \times N}$  the covariance matrix of asset returns. In this framework, the expected return of the portfolio is  $w^T \mu$ , while the portfolio variance  $w^T \Sigma w$  is used as a proxy for risk.

**Risk–Return Trade-Off.** The fundamental idea of MPT is the trade-off between risk and return. A higher expected return is usually associated with higher risk, while reducing risk leads to lower return. This is the main trade-off behind the mean-variance portfolio model discussed in this section. In this sense, the objective is to find a balanced portfolio: either the highest expected return for a given level of risk, or the lowest risk for a given target return.

The mean-variance optimization problem is given by:

$$\max_w \quad w^T \mu - \frac{\lambda}{2} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{1}^T w = 1, \quad w \geq 0,$$

where the parameter  $\lambda > 0$  controls the degree of risk aversion: a larger value of  $\lambda$  gives more weight to risk reduction, whereas a smaller value gives more importance to maximizing expected return. The constraint  $\mathbf{1}^T w = 1$  means that the full budget is invested, and the constraint  $w \geq 0$  excludes short-selling.

**Interpretation of the model.** From an optimization point of view, the problem is a quadratic program. Indeed, the objective function is quadratic in the portfolio weights because of the variance term  $w^T \Sigma w$ , while the constraints  $\mathbf{1}^T w = 1$  and  $w \geq 0$  are linear. For this reason, the formulation remains tractable and can be solved efficiently with standard quadratic optimization methods.

Although portfolio risk is often interpreted through volatility, i.e. the standard deviation of returns, it is more convenient to work with the variance in the optimization problem. Since the square root introduces additional nonlinearity, the variance leads to a simpler formulation while keeping the same interpretation of risk.

An important feature of the model is that each value of  $\lambda$  gives one optimal portfolio, denoted by  $w^*(\lambda)$ . When  $\lambda$  changes, the model produces different optimal portfolios with different risk-return combinations. Together, these portfolios form the efficient frontier. When  $\lambda$  is small, the investor places more emphasis on return, leading to more aggressive portfolios. When  $\lambda$  is large, risk reduction becomes more important, leading to more conservative portfolios. In the limiting case  $\lambda = 0$ , the formulation reduces to the global maximum return portfolio, while for very large  $\lambda$  it approaches the global minimum variance portfolio.

If the non-negativity constraint  $w \geq 0$  is temporarily ignored, the solution can be derived analytically from the first-order optimality conditions. In that case, one introduces a Lagrange multiplier  $\nu$  for the budget constraint  $\mathbf{1}^T w = 1$  and considers the Lagrangian

$$\mathcal{L}(w, \nu) = w^T \mu - \frac{\lambda}{2} w^T \Sigma w + \nu(\mathbf{1}^T w - 1).$$

Setting the gradient with respect to  $w$  equal to zero yields

$$\mu - \lambda \Sigma w + \nu \mathbf{1} = 0,$$

and therefore

$$w^* = \frac{1}{\lambda} \Sigma^{-1} (\mu + \nu \mathbf{1}).$$

Using the budget constraint  $\mathbf{1}^T w^* = 1$ , we obtain

$$\mathbf{1}^T w^* = \frac{1}{\lambda} (\mathbf{1}^T \Sigma^{-1} \mu + \nu \mathbf{1}^T \Sigma^{-1} \mathbf{1}) = 1.$$

Hence,

$$\nu = \frac{\lambda - \mathbf{1}^T \Sigma^{-1} \mu}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}.$$

Substituting this value back gives the closed-form solution

$$w^* = \frac{1}{\lambda} \Sigma^{-1} \left( \mu + \frac{\lambda - \mathbf{1}^T \Sigma^{-1} \mu}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}} \mathbf{1} \right).$$

This final expression no longer contains the Lagrange multiplier explicitly. It shows how the optimal weights depend on the expected return vector, the covariance matrix, and the investor's risk aversion. This quadratic structure makes the model easier to analyze and solve. The derivation above corresponds to the budget-only case, where the non-negativity constraint is not imposed. If additional linear constraints are included, the same Lagrangian idea can still be used, but extra multipliers are needed.

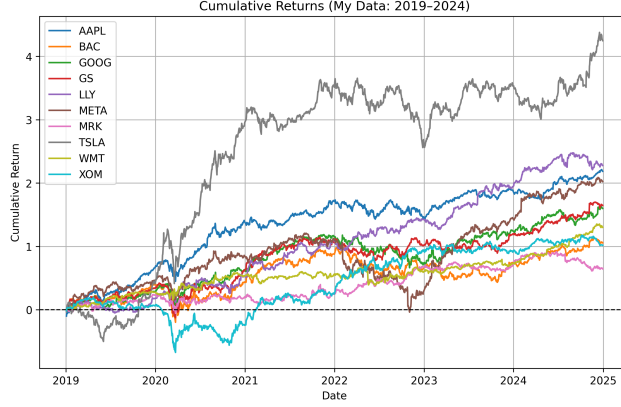


Figure 1: Cumulative returns of selected stocks during the modelling period 2019–2024.

**Data description.** We use daily adjusted close prices for ten large-cap US stocks: AAPL, BAC, GOOG, GS, LLY, META, MRK, TSLA, WMT, and XOM. The data are taken from Yahoo Finance over the modelling period from January 2019 to December 2024. Daily returns are computed for each stock, and Figure 1 shows the cumulative returns during this period. The period from January 2019 to December 2024 is used for parameter estimation, while January 2025 is kept as the out-of-sample backtesting period.

**Interpretation of Figure 1.** The figure presents the cumulative returns of the selected stocks over the modelling period (2019–2024). We observe clear differences in performance across assets. For example, TSLA shows the strongest cumulative growth during the modelling period, while several other stocks have more moderate cumulative returns. The figure also shows that the cumulative return paths fluctuate differently across stocks. These observations highlight the heterogeneity in stock performance, which motivates the use of portfolio optimization to improve diversification and risk management.

**Limitations of Mean-Variance.** Despite its simplicity, the mean-variance model has important limitations. First, variance penalizes gains and losses equally, while investors are mainly concerned with downside risk and usually prefer upside movements of returns. Second, variance does not capture extreme events in the tails of the return distribution. Third, the model is highly sensitive to estimation errors in  $\mu$  and  $\Sigma$ , which may lead to unstable portfolios. Moreover, if only simple constraints are used, the model may produce highly concentrated portfolios, where a large part of the capital is allocated to only a few assets. To measure portfolio concentration, we also use the Herfindahl–

Hirschman Index (HHI), defined by

$$\text{HHI}(w) = \sum_{i=1}^N w_i^2.$$

A larger HHI indicates a more concentrated portfolio, while a smaller HHI indicates a more diversified allocation. For example, if all capital is invested in one asset, then  $\text{HHI} = 1$ .

We now use the same ten-stock daily return data from Figure 1 to illustrate three different behaviours of the mean–variance model.

**Example 1: Mean–variance portfolio with no short-selling.** First, we solve the mean–variance problem with the budget constraint  $\mathbf{1}^T w = 1$  and the no-short-selling constraint  $w \geq 0$ . The result is reported in Table 1. The portfolio is fully concentrated in TSLA, with a maximum weight of 100% and an HHI equal to 1. This shows that the mean–variance model may still produce a highly concentrated portfolio even when short-selling is not allowed.

**Example 2: Mean–variance portfolio with a box constraint.** Next, we add the upper-bound constraint  $0 \leq w_i \leq 0.15$ . This is called a box constraint because each portfolio weight is restricted to stay inside a fixed interval. The result is also reported in Table 1. In this case, the portfolio becomes more diversified: AAPL, GOOG, GS, LLY, META, and TSLA each receive a weight of 15%, WMT receives 10%, and BAC, MRK, and XOM receive zero weight. The HHI decreases from 1.0000 to 0.1450.

| Model            | Expected Return | Volatility | Max Weight | Selected Assets | HHI    |
|------------------|-----------------|------------|------------|-----------------|--------|
| No short-selling | 0.28%           | 4.06%      | 100%       | 1               | 1.0000 |
| Box-constrained  | 0.15%           | 1.58%      | 15%        | 7               | 0.1450 |

Table 1: Comparison of concentration measures for the no-short-selling and box-constrained mean–variance portfolios using the ten-stock daily return data.

**Example 3: Unconstrained mean–variance portfolio.** Finally, we remove the non-negativity constraint and solve the same mean–variance problem using only the budget constraint  $\mathbf{1}^T w = 1$ . The result in Table 2 shows that the unconstrained model produces several extreme negative weights and very large positive weights.

| Quantity                   | Value   |
|----------------------------|---------|
| Minimum weight             | -7.1263 |
| Maximum weight             | 5.8558  |
| Number of negative weights | 5       |
| Sum of weights             | 1.0000  |

Table 2: Summary of the unconstrained mean–variance portfolio weights.



Although the weights still sum to one, this portfolio implies excessive leverage because it combines very large long and short positions. This is risky in practice, since small changes in asset returns can lead to large changes in portfolio value. Therefore, risk-averse investors usually impose constraints such as no short-selling and upper bounds on portfolio weights.

Another important limitation is that diversification depends strongly on the correlation structure between assets, as captured by the covariance matrix  $\Sigma$ . When correlations are low, losses in one asset may be partly offset by gains in others, reducing overall portfolio risk. However, in practice, correlations are not constant and tend to increase during periods of market stress. In such situations, assets may move together, leading to a reduction in diversification benefits. As a result, the mean-variance framework may underestimate portfolio risk precisely when diversification is most needed.

**Motivation for Improvement.** These limitations motivate improvements of the mean-variance framework. Two main directions are commonly considered:

- (i) using alternative risk measures that better capture downside and tail risk;
- (ii) adding practical constraints to control portfolio weights and improve diversification.

We present these two approaches in the following subsections.

### 3.2 Portfolio Optimization with Different Risk Measures

To overcome the limitations of variance, alternative risk measures have been introduced. Let  $R_{p,t}(w)$  denote the portfolio return and define the portfolio loss by

$$L_{p,t}(w) = -R_{p,t}(w).$$

Alternative risk measures can then be written as functions of the portfolio loss instead of the variance.

**Downside Risk (Semi-Variance).** A common downside risk measure is semi-variance, which focuses only on unfavorable outcomes. If  $R_{p,t}(w)$  denotes the portfolio return at time  $t$ , then semi-variance is defined by

$$\text{SemiVar}(w) = \mathbb{E} [\min(R_{p,t}(w), 0)^2].$$

Since  $R_{p,t}(w) = w^T r_t$ , this can also be written as

$$\text{SemiVar}(w) = \mathbb{E} [\min(w^T r_t, 0)^2].$$

**Value-at-Risk (VaR).** Value-at-Risk (VaR) at confidence level  $\alpha$  is defined as:

$$\text{VaR}_\alpha(L_{p,t}(w)) = \inf\{x \in \mathbb{R} : \mathbb{P}(L_{p,t}(w) \leq x) \geq \alpha\}.$$

**Conditional Value-at-Risk (CVaR).** CVaR measures the expected loss beyond VaR:

Conditional Value-at-Risk (CVaR) improves VaR by measuring the expected loss in the worst cases:

$$\text{CVaR}_\alpha(L_{p,t}(w)) = \mathbb{E}[L_{p,t}(w) \mid L_{p,t}(w) \geq \text{VaR}_\alpha(L_{p,t}(w))].$$

Equivalently, CVaR can also be represented as the average of VaR levels beyond  $\alpha$ :

$$\text{CVaR}_\alpha(L_{p,t}(w)) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_\beta(L_{p,t}(w)) d\beta.$$

**Remark (Drawdown-Based Measures).** Drawdown measures consider the decline of portfolio value from a peak. They capture cumulative losses over time and provide a more realistic view of risk in practice.

### 3.3 Portfolio Optimization with Different Constraints

Another way to improve the classical mean-variance model is to keep the same objective function and modify the feasible set by adding practical constraints. In real portfolio management, investors usually face trading limits, diversification requirements, leverage control, and transaction cost considerations. For this reason, the portfolio weights are often restricted by additional constraints.

A general constrained mean-variance problem can be written as

$$\max_w \quad w^T \mu - \frac{\lambda}{2} w^T \Sigma w$$

subject to

$$w \in \mathcal{W},$$

where  $\mathcal{W}$  denotes the feasible set defined by the chosen constraints.

Typical practical constraints include:

1. No short-selling:

$$w \geq 0.$$

2. Leverage control:

$$\|w\|_1 \leq \gamma.$$

Here,  $\|w\|_1 = \sum_{i=1}^N |w_i|$  denotes the total absolute exposure of the portfolio. This constraint is especially useful when short-selling is allowed, because the budget constraint  $\mathbf{1}^T w = 1$  alone does not control this total exposure. As shown in Example 3, the budget constraint alone may allow very large long and short positions. The leverage constraint  $\|w\|_1 \leq \gamma$  limits this exposure and helps prevent unrealistic long-short portfolios.

3. Upper bound constraint:

$$w_i \leq u$$

for upper bounds on individual positions. Together with the no-short-selling constraint  $w_i \geq 0$ , this gives the long-only box constraint

$$0 \leq w_i \leq u.$$

This is the type of constraint used in the box-constrained portfolio example 2.

4. Turnover control:

$$\|w - w_0\|_1 \leq \tau.$$

Here,  $w_0$  denotes the currently held portfolio before rebalancing and  $\tau > 0$  is the maximum allowed turnover. Since

$$\|w - w_0\|_1 = \sum_{i=1}^N |w_i - w_{0,i}|,$$

it measures the total absolute adjustment in portfolio weights. Therefore, a smaller value of  $\tau$  forces the new portfolio to stay closer to the current portfolio and reduces the amount of buying and selling required. This is useful in practice because frequent or large rebalancing creates transaction costs.

5. Sparsity or cardinality control:

$$\|w\|_0 \leq K$$

for sparsity or cardinality control, where  $\|w\|_0$  denotes the number of nonzero portfolio weights and  $K$  is the maximum number of active positions. This constraint limits the number of assets included in the portfolio. For example, if  $K = 5$ , then at most five assets can receive nonzero weights. Such a constraint may be useful when an investor wants a simpler portfolio with fewer positions to monitor and rebalance. However, the  $\ell_0$ -constraint is nonconvex and is therefore more difficult to handle computationally.

These additional constraints make the portfolio more realistic in practice. They can reduce concentration, control leverage, limit rebalancing costs, and improve diversification.

**Example 4: Turnover constraint.** To illustrate the effect of the turnover constraint, we add a small empirical example. The current portfolio  $w_0$  is taken to be the equal-weight portfolio, and we impose the turnover limit

$$\|w - w_0\|_1 \leq 0.30.$$

This means that the total absolute change from the current portfolio to the new portfolio cannot exceed 30%. Table 3 compares the box-constrained portfolio with the portfolio obtained after adding this turnover constraint.

| Model                     | Expected Return | Volatility | Max Weight | Selected Assets | HHI    | Turnover from $w_0$ |
|---------------------------|-----------------|------------|------------|-----------------|--------|---------------------|
| Box-constrained           | 0.15%           | 1.58%      | 15.00%     | 7               | 0.1450 | 0.4000              |
| Box + turnover constraint | 0.14%           | 1.49%      | 15.00%     | 9               | 0.1200 | 0.3000              |

Table 3: Effect of adding a turnover constraint to the box-constrained portfolio.

The results show that the original box-constrained portfolio requires a turnover of 0.4000 relative to the equal-weight portfolio. After adding the turnover constraint with  $\tau = 0.30$ , the turnover is reduced to 0.3000, exactly satisfying the imposed limit. The expected return decreases slightly from 0.15% to 0.14%, but the volatility also decreases from 1.58% to 1.49%. Moreover, the number of selected assets increases from 7 to 9 and the HHI decreases from 0.1450 to 0.1200. This shows that the turnover constraint keeps the new portfolio closer to the current allocation and in this empirical example, it also leads to a more diversified allocation, as the number of selected assets increases and the HHI decreases.

## 4 Risk Parity Portfolios

Risk parity is an alternative portfolio construction approach that focuses on allocating risk rather than capital. In the classical mean–variance framework, portfolio choice is based on the trade-off between expected return and variance. However, this approach does not explicitly show how much each asset contributes to the total portfolio risk. Risk parity addresses this issue by constructing a portfolio in which the assets contribute to total risk in a balanced way.

To define this idea, we first choose a portfolio risk measure. In this section, we use portfolio volatility, defined by

$$\sigma(w) = \sqrt{w^T \Sigma w},$$

The total portfolio risk can be decomposed into the contributions of the individual assets. The marginal risk contribution of asset  $i$  is defined as the sensitivity of portfolio volatility with respect to the weight  $w_i$ :

$$\text{MRC}_i(w) = \frac{\partial \sigma(w)}{\partial w_i}.$$

The risk contribution of asset  $i$  is then defined by

$$\text{RC}_i(w) = w_i \text{MRC}_i(w).$$

Since volatility is a homogeneous function of degree one, Euler’s theorem gives the decomposition

$$\sigma(w) = \sum_{i=1}^N \text{RC}_i(w).$$

Therefore, the relative risk contribution of asset  $i$  is defined as

$$\text{RRC}_i(w) = \frac{\text{RC}_i(w)}{\sigma(w)}, \quad \sum_{i=1}^N \text{RRC}_i(w) = 1.$$

For the volatility risk measure, these quantities have explicit forms. Since

$$\sigma(w) = \sqrt{w^T \Sigma w},$$

we obtain

$$\text{MRC}_i(w) = \frac{(\Sigma w)_i}{\sqrt{w^T \Sigma w}},$$

and hence

$$\text{RC}_i(w) = w_i \frac{(\Sigma w)_i}{\sqrt{w^T \Sigma w}}.$$

It follows that

$$\text{RRC}_i(w) = \frac{w_i (\Sigma w)_i}{w^T \Sigma w}.$$

A risk parity portfolio is obtained when all assets have the same relative risk contribution:

$$\text{RRC}_i(w) = \frac{1}{N}, \quad i = 1, \dots, N.$$

Equivalently, the risk-parity weight vector  $w^{\text{RP}}$  is obtained by solving

$$\text{RRC}_i(w^{\text{RP}}) = \frac{1}{N}, \quad i = 1, \dots, N,$$

together with the chosen portfolio constraints. Thus, risk parity does not require equal capital weights. Instead, it requires equal contributions to total portfolio risk.

More generally, one can consider a risk budgeting portfolio. In this case, each asset is assigned a target risk budget  $b_i$ , where

$$b_i \geq 0, \quad \sum_{i=1}^N b_i = 1.$$

The target condition becomes

$$\text{RRC}_i(w) = b_i, \quad i = 1, \dots, N.$$

Risk parity is therefore the special case of risk budgeting with

$$b_i = \frac{1}{N}, \quad i = 1, \dots, N.$$

Using the expression of the relative risk contribution under volatility, the risk budgeting condition can be written as

$$\frac{w_i (\Sigma w)_i}{w^T \Sigma w} = b_i, \quad i = 1, \dots, N.$$

Equivalently,

$$w_i(\Sigma w)_i = b_i w^T \Sigma w, \quad i = 1, \dots, N.$$

In practice, exact equality may be difficult to satisfy, especially when additional constraints are imposed. Therefore, the risk budgeting problem is often written as an optimization problem:

$$\min_w \sum_{i=1}^N (\text{RRC}_i(w) - b_i)^2$$

subject to

$$w \in \mathcal{W},$$

where  $\mathcal{W}$  denotes the feasible set of portfolio weights. For example,  $\mathcal{W}$  may include the budget constraint  $1^T w = 1$  and the no-short-selling constraint  $w \geq 0$ . This formulation shows the main difference between equal-weight and risk-parity portfolios. The equal-weight portfolio allocates the same capital to each asset, whereas the risk-parity portfolio allocates risk more evenly across assets. For this reason, risk parity can be useful when the objective is to improve diversification from a risk perspective.

## 5 Portfolio Backtesting

Backtesting is used to evaluate how the optimized portfolios perform on out-of-sample data, that is, data not used to estimate the model parameters. In this project, the portfolio weights are computed using the modelling period described in the data description, while their performance is evaluated over a separate out-of-sample period. This allows us to check whether the optimized portfolios remain stable and useful beyond the sample used for optimization.

We compare the following portfolios:

1. the no-short-selling mean-variance portfolio;
2. the box-constrained mean-variance portfolio with  $0 \leq w_i \leq 0.15$ ;
3. the equal-weight benchmark, where all assets receive the same weight.

All performance measures are computed from daily returns; in particular, the Sharpe ratio reported below is a daily Sharpe ratio and is not annualized.

We evaluate the strategies using total return, daily mean return, daily volatility, daily Sharpe ratio, and maximum drawdown. The daily Sharpe ratio is computed as

$$\text{Daily Sharpe Ratio} = \frac{\text{Daily Mean Return}}{\text{Daily Volatility}}.$$

It measures the average return obtained per unit of daily risk. All quantities in Table 4 are based on daily returns and are not annualized.

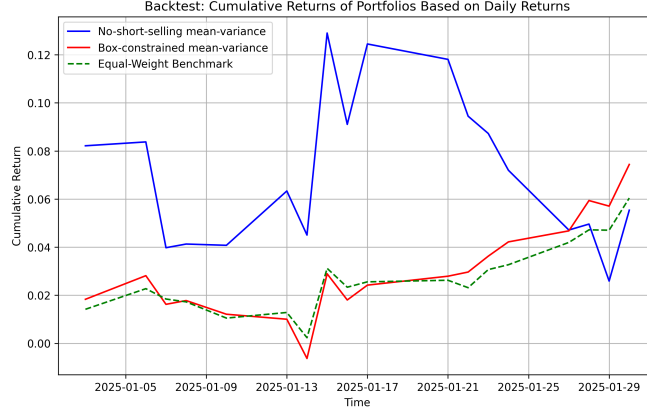


Figure 2: Out-of-sample cumulative returns of the optimized portfolios and the equal-weight benchmark based on daily returns.

| Strategy                       | Total Return | Daily Mean Return | Daily Volatility | Daily Sharpe Ratio | Max Drawdown |
|--------------------------------|--------------|-------------------|------------------|--------------------|--------------|
| No-short-selling mean-variance | 5.54%        | 0.35%             | 3.34%            | 0.11               | -9.14%       |
| Box-constrained mean-variance  | 7.44%        | 0.41%             | 1.18%            | 0.34               | -3.35%       |
| Equal-Weight Benchmark         | 6.04%        | 0.33%             | 0.91%            | 0.36               | -1.99%       |

Table 4: Out-of-sample backtesting performance of the optimized portfolios and the equal-weight benchmark based on daily returns.

The backtest results show that the no-short-selling portfolio has a positive total return, but it also has the highest daily volatility and the largest maximum drawdown. This is consistent with the previous concentration analysis, where this portfolio allocated all capital to only one asset. The box-constrained mean-variance portfolio with  $0 \leq w_i \leq 0.15$  achieves the highest total return over the backtesting period. It also has much lower volatility and a smaller drawdown than the no-short-selling mean-variance portfolio. Compared with the equal-weight benchmark, it gives a higher total return, although the benchmark has a slightly higher daily Sharpe ratio and a lower maximum drawdown. Overall, this empirical example suggests that adding the upper-bound constraint improves the practical behaviour of the mean-variance portfolio. It reduces concentration risk and gives a more stable out-of-sample performance relative to the no-short-selling mean-variance portfolio. However, the benchmark still performs well in terms of Sharpe ratio and drawdown, which shows that backtesting should be interpreted as a validation tool rather than as a proof of future performance.

## 6 Conclusion

The main point of this report is that the classical mean-variance model is a useful starting point, because it explains the trade-off between expected return and risk in a clear way. However, the examples show that the basic model alone

is not always realistic. The no-short-selling portfolio became fully concentrated in one stock, while the unconstrained portfolio produced very large long and short positions.

Adding practical constraints improved the behaviour of the portfolio. In particular, the box constraint reduced concentration and gave a more stable out-of-sample performance in the backtest. Still, the equal-weight benchmark also performed reasonably well, which shows that optimization does not automatically mean better performance.

Overall, the final portfolio depends on the choice of risk measure, constraints, and backtesting setup. With more time, this analysis could be extended to other models, such as higher-order portfolio models, and to longer backtesting periods.

## References

- [1] Markowitz, H. M. (1952). *Portfolio Selection. The Journal of Finance*, 7(1), 77–91.
- [2] Palomar, D. P. (2025). *Portfolio Optimization: Theory and Application*. Cambridge University Press.